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APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
18/501,015	11/02/2023	Hsiang-Feng Chi	124536-US-PA	3410

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EXAMINER	
RIDDER, CLAYTON PAUL	

ART UNIT	PAPER NUMBER
3646	

NOTIFICATION DATE	DELIVERY MODE
09/30/2025	ELECTRONIC

Please find below and/or attached an Office communication concerning this application or proceeding.

The time period for reply, if any, is set in the attached communication.

Notice of the Office communication was sent electronically on above-indicated "Notification Date" to the following e-mail address(es):

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Office Action Summary	Application No. 18/501,015	Applicant(s) Chi, Hsiang-Feng	
	Examiner CLAYTON P RIDDER	Art Unit 3646	AIA (First Inventor to File) Status Yes

-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address --

Period for Reply

A SHORTENED STATUTORY PERIOD FOR REPLY IS SET TO EXPIRE 3 MONTHS FROM THE MAILING DATE OF THIS COMMUNICATION.

- Extensions of time may be available under the provisions of 37 CFR 1.136(a). In no event, however, may a reply be timely filed after SIX (6) MONTHS from the mailing date of this communication.
- If NO period for reply is specified above, the maximum statutory period will apply and will expire SIX (6) MONTHS from the mailing date of this communication.
- Failure to reply within the set or extended period for reply will, by statute, cause the application to become ABANDONED (35 U.S.C. § 133). Any reply received by the Office later than three months after the mailing date of this communication, even if timely filed, may reduce any earned patent term adjustment. See 37 CFR 1.704(b).

Status

1)☑ Responsive to communication(s) filed on 11/02/2023.

☐ A declaration(s)/affidavit(s) under **37 CFR 1.130(b)** was/were filed on _____.

2a)☐ This action is **FINAL.** 2b) ☑ This action is non-final.

3)☐ An election was made by the applicant in response to a restriction requirement set forth during the interview on _____; the restriction requirement and election have been incorporated into this action.

4)☐ Since this application is in condition for allowance except for formal matters, prosecution as to the merits is closed in accordance with the practice under *Ex parte Quayle*, 1935 C.D. 11, 453 O.G. 213.

Disposition of Claims*

5) ☑ Claim(s) 1-19 is/are pending in the application.

5a) Of the above claim(s) _____ is/are withdrawn from consideration.

6) ☐ Claim(s) _____ is/are allowed.

7) ☑ Claim(s) 1-19 is/are rejected.

8) ☐ Claim(s) _____ is/are objected to.

9) ☐ Claim(s) _____ are subject to restriction and/or election requirement

* If any claims have been determined allowable, you may be eligible to benefit from the **Patent Prosecution Highway** program at a participating intellectual property office for the corresponding application. For more information, please see http://www.uspto.gov/patents/init_events/pph/index.jsp or send an inquiry to **PPHfeedback@uspto.gov.**

Application Papers

10)☐ The specification is objected to by the Examiner.

11)☑ The drawing(s) filed on 11/02/2023 is/are: a)☑ accepted or b)☐ objected to by the Examiner.

Applicant may not request that any objection to the drawing(s) be held in abeyance. See 37 CFR 1.85(a).

Replacement drawing sheet(s) including the correction is required if the drawing(s) is objected to. See 37 CFR 1.121(d).

Priority under 35 U.S.C. § 119

12)☑ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).

Certified copies:

a)☑ All b)☐ Some** c)☐ None of the:

1.☑ Certified copies of the priority documents have been received.

2.☐ Certified copies of the priority documents have been received in Application No. _____.

3.☐ Copies of the certified copies of the priority documents have been received in this National Stage application from the International Bureau (PCT Rule 17.2(a)).

** See the attached detailed Office action for a list of the certified copies not received.

Attachment(s)

1) ☑ Notice of References Cited (PTO-892)

2) ☑ Information Disclosure Statement(s) (PTO/SB/08a and/or PTO/SB/08b) Paper No(s)/Mail Date _____.

3) ☐ Interview Summary (PTO-413) Paper No(s)/Mail Date _____.

4) ☐ Other: _____.

DETAILED ACTION

Notice of Pre-AIA or AIA Status

1. The present application, filed on or after March 16, 2013, is being examined under the first inventor to file provisions of the AIA.

2. In the event the determination of the status of the application as subject to AIA 35 U.S.C. 102 and 103 (or as subject to pre-AIA 35 U.S.C. 102 and 103) is incorrect, any correction of the statutory basis (i.e., changing from AIA to pre-AIA) for the rejection will not be considered a new ground of rejection if the prior art relied upon, and the rationale supporting the rejection, would be the same under either status.

Claim Rejections - 35 USC § 102

3. The following is a quotation of the appropriate paragraphs of 35 U.S.C. 102 that form the basis for the rejections under this section made in this Office action:

> A person shall be entitled to a patent unless –
>
> (a)(1) the claimed invention was patented, described in a printed publication, or in public use, on sale, or otherwise available to the public before the effective filing date of the claimed invention.
>
> (a)(2) the claimed invention was described in a patent issued under section 151, or in an application for patent published or deemed published under section 122(b), in which the patent or application, as the case may be, names another inventor and was effectively filed before the effective filing date of the claimed invention.

4. **Claims 1. 4-9, 11, 12, 15-19** are rejected under 35 U.S.C. 102(a)(1) as being anticipated by **LAGHEZZA(CN111880169A)**.

Regarding claim 1, LAGHEZZA discloses

A radar apparatus, comprising: a frequency synthesizer, configured to generate a carrier signal **("PLL111 generates ramp FMCW radar signal" [0049])**, wherein a frequency of the carrier signal changes with time within a sweep period of the carrier signal **("the transmitting-side circuit generates a radar waveform sequence by sweeping a carrier signal over a frequency range" [0045])**, and the carrier

signal comprises a frequency raising section and a frequency decreasing section **("an FMCW signal that may be transmitted as a ramped radar chirp" [0040])**; a signal generator, configured to generate a shaping signal **(" the PSK input can be provided by the reduced angle PSK (Mr-PSK) module 113." [0049])**, wherein the shaping signal comprises a first section and a second section, the first section and the second section have different amplitudes, and the second section corresponds to a turn-around section between the frequency raising section and the frequency decreasing section **("Rectangular pulse shaping can be used so that [the communication signal] is constant for the duration" [0050])**; and a transmitting front-end circuit, coupled to the frequency synthesizer and the signal generator and configured to generate a transmitting signal according to the carrier signal and the shaping signal **(FIG.1, Part 115)**

Regarding claim 4, LAGHEZZA discloses

The radar apparatus according to claim 1, wherein an amplitude of the second section is smaller than the first section **("Rectangular pulse shaping can be used so that [...] is constant for the duration" [0050] The examiner notes that the claimed difference in shaping signal amplitude is implicit to a rectangular wave. All rectangular waves include a period of maximum amplitude, a descending period and an ascending period.).**

Regarding claim 5, LAGHEZZA discloses

The radar apparatus according to claim 1, wherein a waveform of the second section is defined by a sin-squared function, a linear function, or a rectangular function **("Rectangular pulse shaping can be used so that [...] is constant for the duration" [0050]).**

Regarding claim 6, LAGHEZZA discloses

The radar apparatus according to claim 1, wherein an amplitude is the same during the first section at a time domain **("Rectangular pulse shaping can be used so that [...] is constant for the duration" [0050] The examiner notes that the claimed constant amplitude is implicit to a rectangular wave).**

Regarding claim 7, LAGHEZZA discloses

The radar apparatus according to claim 1, wherein the transmitting front-end circuit comprises: a first mixer, coupled to the signal generator and the frequency synthesizer **(FIG.1, Part.112)** and configured to mix the carrier signal and the shaping signal to generate a mixed signa **("The mixer 112 uses the PSK signal from the Mr-PSK module 113 to mix the ramp FMCW"[0050])**l; and a first amplifier, coupled to a first antenna and configured to amplify the mixed signal to generate the transmitting signal **("power amplifier 114" [0049])**, and the radar apparatus further comprising: the first antenna, coupled to the first amplifier and configured to emit the transmitting signal **("The power amplifier 114 amplifies the output of the mixer 112 to be transmitted via the antenna 115" [0049]).**

Regarding claim 8, LAGHEZZA discloses

The radar apparatus as according to claim 1, further comprising: a controller, coupled to the signal generator and the frequency synthesizer and configured to synchronize the second section to the turn-around section between the frequency raising section and the frequency decreasing section **("The transmitting-side circuit 110 and/or the receiving-side circuit 120 may be synchronized using the timing block 130 ." [0050])**

Regarding claim 9, LAGHEZZA discloses

The radar apparatus according to claim 1, wherein the signal generator comprises: a wave generator, configured to generate an initial shaping signal **("the transmit-side circuit includes an oscillator"**

[0045]); and a digital to analog converter, coupled to the wave generator and configured to convert the initial shaping signal into the shaping signal **("the transmit-side circuitry includes a digital signal processing circuit for processing a digitized version of the communication data and a mixer for performing frequency up-conversion." [0042])**

Regarding claim 11, LAGHEZZA discloses

A signal processing circuit, comprising: a frequency synthesizer, configured to generate a carrier signal **("PLL111 generates ramp FMCW radar signal" [0049])**, wherein a frequency of the carrier signal changes with time within a sweep period of the carrier signal **("the transmitting-side circuit generates a radar waveform sequence by sweeping a carrier signal over a frequency range" [0045])**, and the carrier signal comprises a frequency raising section and a frequency decreasing section **("an FMCW signal that may be transmitted as a ramped radar chirp" [0040])**; and a signal generator, configured to generate a shaping signal, wherein the shaping signal comprises a first section and a second section, the first section and the second section have different amplitudes, and the second section corresponds to a turn-around section between the frequency raising section and the frequency decreasing section **(" the PSK input can be provided by the reduced angle PSK (Mr-PSK) module 113." [0049]).**

Regarding claim 12, LAGHEZZA discloses

A signal processing method for a radar apparatus, comprising: generating a carrier signal **("PLL111 generates ramp FMCW radar signal" [0049])**, wherein a frequency of the carrier signal changes with time within a sweep period of the carrier signal **("the transmitting-side circuit generates a radar waveform sequence by sweeping a carrier signal over a frequency range" [0045])**, and the carrier signal comprises a frequency raising section and a frequency decreasing section **("an FMCW signal that may be transmitted as a ramped radar chirp" [0040])**; generating a shaping signal **(" the PSK input can**

be provided by the reduced angle PSK (Mr-PSK) module 113." [0049]), wherein the shaping signal comprises a first section and a second section, the first section and the second section have different amplitudes, and the second section corresponds to a turn-around section between the frequency raising section and the frequency decreasing section **("Rectangular pulse shaping can be used so that [the communication signal] is constant for the duration" [0050])**; and generating a transmitting signal according to the carrier signal and the shaping signal, wherein the transmitting signal is configured to allow the radar apparatus to emit **(FIG.1, Part 115)**

Regarding claim 15, LAGHEZZA discloses

The signal processing method for the radar apparatus according to claim 12, wherein an amplitude of the second section is smaller than the first section **("Rectangular pulse shaping can be used so that [...] is constant for the duration" [0050] The examiner notes that the claimed difference in shaping signal amplitude is implicit to a rectangular wave. All rectangular waves include a period of maximum amplitude, a descending period and an ascending period).**

Regarding claim 16, LAGHEZZA discloses

The signal processing method for the radar apparatus according to claim 12, wherein a waveform of the second section is defined by a sin-squared function, a linear function, or a rectangular function **("Rectangular pulse shaping can be used so that [...] is constant for the duration" [0050]).**

Regarding claim 17, LAGHEZZA discloses

The signal processing method for the radar apparatus according to claim 12, wherein an amplitude is the same during the first section at a time domain **("Rectangular pulse shaping can be used so that [...] is**

constant for the duration" [0050] The examiner notes that the claimed constant amplitude is implicit to a rectangular wave).

Regarding claim 18, LAGHEZZA discloses

The signal processing method for the radar apparatus according to claim 12, further comprising: synchronizing the second section with the turn-around section between the frequency raising section and the frequency decreasing section **("The transmitting-side circuit 110 and/or the receiving-side circuit 120 may be synchronized using the timing block 130 ." [0050]).**

Regarding claim 19, LAGHEZZA discloses

The signal processing method for the radar apparatus according to claim 12, further comprising: generating an initial shaping signal **(" the PSK input can be provided by the reduced angle PSK (Mr-PSK) module 113." [0049])**; and converting the initial shaping signal from a digital form to an analog form of the shaping signal **("the transmit-side circuitry includes a digital signal processing circuit for processing a digitized version of the communication data and a mixer for performing frequency up-conversion." [0042]).**

Claim Rejections - 35 USC § 103

5. The following is a quotation of 35 U.S.C. 103 which forms the basis for all obviousness rejections set forth in this Office action:

> A patent for a claimed invention may not be obtained, notwithstanding that the claimed invention is not identically disclosed as set forth in section 102, if the differences between the claimed invention and the prior art are such that the claimed invention as a whole would have been obvious before the effective filing date of the claimed invention to a person having ordinary skill in the art to which the claimed invention pertains. Patentability shall not be negated by the manner in which the invention was made.

6. The factual inquiries for establishing a background for determining obviousness under 35 U.S.C. 103 are summarized as follows:

1. Determining the scope and contents of the prior art.

2. Ascertaining the differences between the prior art and the claims at issue.

3. Resolving the level of ordinary skill in the pertinent art.

4. Considering objective evidence present in the application indicating obviousness or nonobviousness.

7. **Claims 2, 3, 13, and 14** are rejected under 35 U.S.C. 103 as being unpatentable over **LAGHEZZA(CN111880169A)** in view of **ELFERINK(EP2597484A1).**

Regarding claim 2, LAGHEZZA discloses all the limitations of claim 1. LAGHEZZA discloses, radar apparatus wherein, the turn-around section corresponds to a junction between the frequency raising section and the frequency decreasing section **("Rectangular pulse shaping can be used so that [the communication signal] is constant for the duration" [0050] The examiner notes that the claimed turn around section is implicit to a periodic rectangular wave.)**

LAGHEZZA discloses a sweep carrier signal but does not explicitly disclose nor limit wherein the wherein the carrier signal is a triangle wave. ELFERINK teaches in the same field of endeavor of carrier signal modulation. ELFERINK discloses the radar apparatus wherein, the carrier signal is a triangle wave **("The radar pulse may, for example, be defined by [...] a triangular frequency-to-time relationship (i.e. one linear up-chirp followed by a corresponding linear down-chirp)" [0009])**

ELFERINK teaches in the same field of endeavor of carrier signal modulation. It would have been obvious to one of ordinary skill in the art prior to the effective filing date of the claimed invention to modify **LAGHEZZA** with the teachings of **ELFERINK** to incorporate the features of a carrier signal being a sawtooth wave so as to gain the advantage of improving range [0021, ELFERINK] . Also, since it has been held that if a technique has been used to improve one device, and a person of ordinary skill in the art

would recognize that it would improve similar devices in the same way, using the technique is obvious unless its actual application is beyond his or her skill (MPEP 2143).

Regarding claim 3, LAGHEZZA discloses all the limitations of claim 1. LAGHEZZA discloses, radar apparatus wherein, and the turn-around section corresponds to an interval from the frequency raising section to the frequency decreasing section **("Rectangular pulse shaping can be used so that [the communication signal] is constant for the duration" [0050] The examiner notes that the claimed turn around section is implicit to a periodic rectangular wave. There is necessarily an interval between the frequency raising section to the frequency decreasing section).**

LAGHEZZA discloses a sweep carrier signal but does not explicitly disclose nor limit wherein the wherein the carrier signal is a sawtooth wave. ELFERINK teaches in the same field of endeavor of carrier signal modulation. ELFERINK discloses the radar apparatus wherein, the carrier signal is a sawtooth wave **("The radar pulse may, for example, be defined by a waveform with a sawtooth-like frequency-to-time relationship (i.e. one linear up- or down chirp) [0009])**

ELFERINK teaches in the same field of endeavor of carrier signal modulation. It would have been obvious to one of ordinary skill in the art prior to the effective filing date of the claimed invention to modify **LAGHEZZA** with the teachings of **ELFERINK** to incorporate the features of a carrier signal being a sawtooth wave so as to gain the advantage of improving range [0021, ELFERINK] . Also, since it has been held that if a technique has been used to improve one device, and a person of ordinary skill in the art would recognize that it would improve similar devices in the same way, using the technique is obvious unless its actual application is beyond his or her skill (MPEP 2143).

Regarding claim 13, LAGHEZZA discloses all the limitations of claim 12. LAGHEZZA discloses, the signal processing method wherein, the turn-around section corresponds to a junction between the frequency raising section and the frequency decreasing section.

LAGHEZZA discloses a sweep carrier signal but does not explicitly disclose nor limit wherein the wherein the carrier signal is a triangle wave. ELFERINK teaches in the same field of endeavor of carrier signal modulation. ELFERINK discloses the signal processing method wherein, the carrier signal is a triangle wave **("The radar pulse may, for example, be defined by [...] a triangular frequency-to-time relationship (i.e. one linear up-chirp followed by a corresponding linear down-chirp)" [0009])**

ELFERINK teaches in the same field of endeavor of carrier signal modulation. It would have been obvious to one of ordinary skill in the art prior to the effective filing date of the claimed invention to modify **LAGHEZZA** with the teachings of **ELFERINK** to incorporate the features of a carrier signal being a sawtooth wave so as to gain the advantage of improving range [0021, ELFERINK] . Also, since it has been held that if a technique has been used to improve one device, and a person of ordinary skill in the art would recognize that it would improve similar devices in the same way, using the technique is obvious unless its actual application is beyond his or her skill (MPEP 2143).

Regarding claim 14, LAGHEZZA discloses all the limitations of claim 12. LAGHEZZA discloses, the signal processing method wherein, and the turn-around section corresponds to an interval from the frequency raising section to the frequency decreasing section **("Rectangular pulse shaping can be used so that [the communication signal] is constant for the duration" [0050] The examiner notes that the claimed turn around section is implicit to a periodic rectangular wave. There is necessarily an interval between the frequency raising section to the frequency decreasing section).**

LAGHEZZA discloses a sweep carrier signal but does not explicitly disclose nor limit wherein the wherein the carrier signal is a sawtooth wave. ELFERINK teaches in the same field of endeavor of

carrier signal modulation. ELFERINK discloses the radar apparatus wherein, the carrier signal is a sawtooth wave **("The radar pulse may, for example, be defined by a waveform with a sawtooth-like frequency-to-time relationship (i.e. one linear up- or down chirp) [0009])**

ELFERINK teaches in the same field of endeavor of carrier signal modulation. It would have been obvious to one of ordinary skill in the art prior to the effective filing date of the claimed invention to modify **LAGHEZZA** with the teachings of **ELFERINK** to incorporate the features of a carrier signal being a sawtooth wave so as to gain the advantage of improving range [0021, ELFERINK] . Also, since it has been held that if a technique has been used to improve one device, and a person of ordinary skill in the art would recognize that it would improve similar devices in the same way, using the technique is obvious unless its actual application is beyond his or her skill (MPEP 2143).

8. **Claim 10** is rejected under 35 U.S.C. 103 as being unpatentable over **LAGHEZZA(CN111880169A)** in view of **ADIB(US20170074980A1)**

Regarding claim 10, LAGHEZZA discloses all the limitations of claim 1. LAGHEZZA does not explicitly disclose nor limit wherein the wherein the radar apparatus is an ultra wideband (UWB) radar. ADIB teaches in the same field of endeavor of carrier signal modulation. ADIB discloses the, radar apparatus wherein, the radar apparatus is an ultra wideband (UWB) radar **("the signal generator is an ultra wide band frequency modulated carrier wave (FMCW) generator" [0079])**

ADIB teaches in the same field of endeavor of carrier signal modulation. It would have been obvious to one of ordinary skill in the art prior to the effective filing date of the claimed invention to modify **LAGHEZZA** with the teachings of **ADIB** to incorporate the features of an ultra wideband (UWB) radar so as to gain the advantage of reducing the effects of noise [0091, ADIB] . Also, since it has been held that if a technique has been used to improve one device, and a person of ordinary skill in the art

would recognize that it would improve similar devices in the same way, using the technique is obvious unless its actual application is beyond his or her skill (MPEP 2143).

9. For applicant's benefit portions of the cited reference(s) have been cited to aid in the review of the rejection(s). While every attempt has been made to be thorough and consistent within the rejection it is noted that the *PRIOR ART MUST BE CONSIDERED IN ITS ENTIRETY, INCLUDING DISCLOSURES THAT TEACH AWAY FROM THE CLAIMS.* See MPEP 2141.02 VI.

Documents Considered but not Relied Upon

10. The prior art made of record and not relied upon is considered pertinent to the applicant's Disclosure.

ANTIDE(US20230082258A1) is considered analogous art to the instant application as it discloses in [0006] "Such a signal, called "chirp", linearly sweeps a band of frequencies B, around a carrier frequency."

Conclusion

11. Any inquiry concerning this communication or earlier communications from the examiner should be directed to CLAYTON PAUL RIDDER whose telephone number is (571)272-2771. The examiner can normally be reached Monday thru Friday ET.

Examiner interviews are available via telephone, in-person, and video conferencing using a USPTO supplied web-based collaboration tool. To schedule an interview, applicant is encouraged to use the USPTO Automated Interview Request (AIR) at http://www.uspto.gov/interviewpractice.

If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, Jack Keith can be reached on (571) 272-6878. The fax phone number for the organization where this application or proceeding is assigned is 571-273-8300.

Information regarding the status of published or unpublished applications may be obtained from Patent Center. Unpublished application information in Patent Center is available to registered users. To file and manage patent submissions in Patent Center, visit: https://patentcenter.uspto.gov. Visit https://www.uspto.gov/patents/apply/patent-center for more information about Patent Center and https://www.uspto.gov/patents/docx for information about filing in DOCX format. For additional questions, contact the Electronic Business Center (EBC) at 866-217-9197 (toll-free). If you would like assistance from a USPTO Customer Service Representative, call 800-786-9199 (IN USA OR CANADA) or 571-272-1000.

/C.P.R./
Examiner, Art Unit 3646

/JACK W KEITH/
Supervisory Patent Examiner, Art Unit 3646